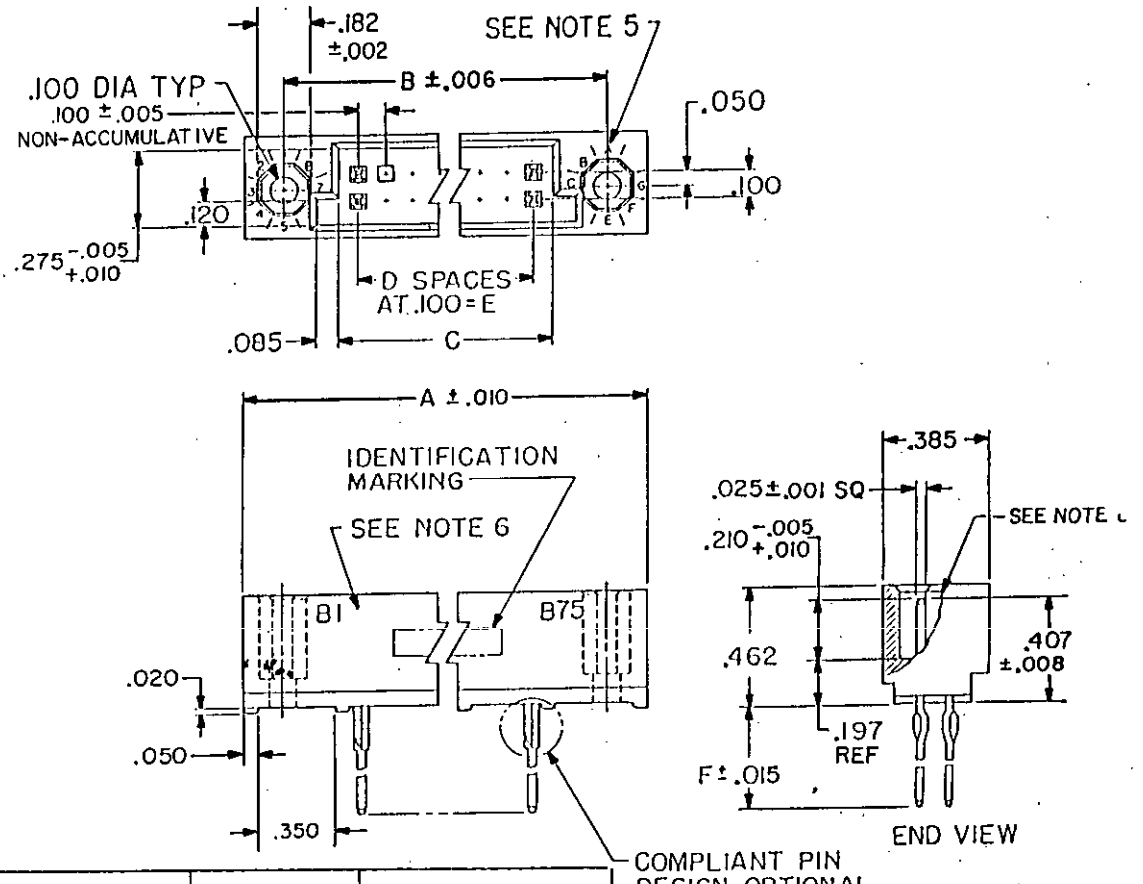
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 2-ROW PIN ASSEMBLIES, 20 THROUGH 150 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

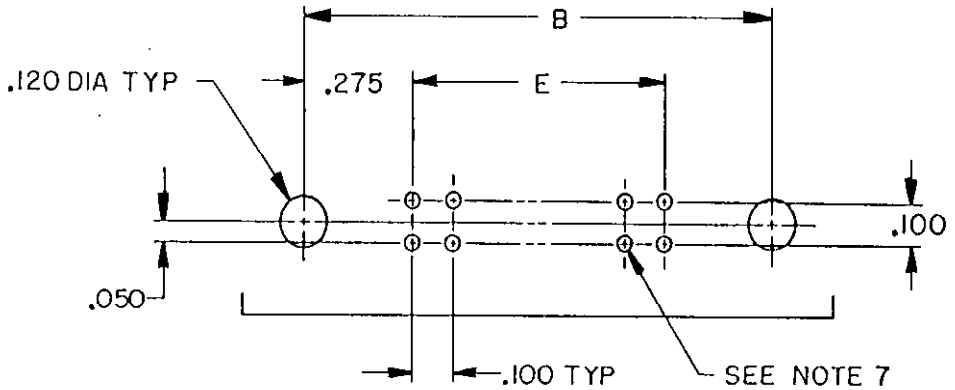


F ±.015		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin
.533	13.54	II	"
.328	8.33	III	"
.250	6.35	IV	"
.190	4.83	V	"
.177	4.50	VI	Solder post 1/
.165	4.19	VII	"
.140	3.56	VIII	"
.120	3.05	IX	"

1/ Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 20 through 150 contacts.

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Recommended circuit layout

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.006	0.15
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.050	1.27
.085	2.16
.100	2.54
.120	3.05
.182	4.62
.190	4.83
.197	5.00
.210	5.33
.250	6.35
.275	6.98
.350	8.89
.380	9.65
.385	9.78
.407	10.34
.462	11.73
.533	13.54
.733	18.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/174.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row closest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042-.050 (1.067-1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ±.0010 (1.15 ±0.0254 mm) drilled hole (1.15 mm drill) .001-.003 (0.025-0.076 mm) thick copper on wall, .0003 (0.0076 mm) min. tin lead plating, finished hole diameter .037-.043 (0.94-1.09 mm) after plating, .036-.043 (0.91-1.09 mm) after reflow, copper hardness (Knoop) 150 maximum.
8. Chamfer shall be .015 minimum at a 45° angle ±5 degrees.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 20 through 150 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions.

Dash no.	No. of contacts	Contact type	A	B	C	D	E	Contact ident. no.
01 02 03 04 05 06 07 08 09	20	I II III IV V VI VII VIII IX	1.750 (44.45)	1.450 (36.83)	1.050 (26.67)	9	.900 (22.86)	1-10
10 11 12 13 14 15 16 17 18	30	I II III IV V VI VII VIII IX	2.250 (57.15)	1.950 (49.53)	1.550 (39.37)	14	1.400 (35.56)	1-15
19 20 21 22 23 24 25 26 27	40	I II III IV V VI VII VIII IX	2.750 (69.85)	2.450 (62.23)	2.050 (52.07)	19	1.900 (48.26)	1-20
28 29 30 31 32 33 34 35 36	50	I II III IV V VI VII VIII IX	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)	1-25
37 38 39 40 41 42 43 44 45	56	I II III IV V VI VII VIII IX	3.550 (90.17)	3.250 (82.55)	2.850 (72.39)	27	2.700 (68.58)	1-28

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Contact type	A	B	C	D	E	Contact ident. no.
46 47 48 49 50 51 52 53 54	60	I II III IV V VI VII VIII IX	3.750 (95.25)	3.450 (87.63)	3.050 (77.47)	29	2.900 (73.66)	1-30
55 56 57 58 59 60 61 62 63	70	I II III IV V VI VII VIII IX	4.250 (107.95)	3.950 (100.33)	3.550 (90.17)	34	3.400 (86.36)	1-35
64 65 66 67 68 69 70 72	80	I II III IV V VI VIII IX	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)	1-40
73 74 75 76 77 78 79 80 81	90	I II III IV V VI VII VIII IX	5.250 (133.35)	4.950 (125.73)	4.550 (115.57)	44	4.400 (111.76)	1-45
82 83 84 85 86 87 88 89 90	100	I II III IV V VI VII VIII IX	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)	1-50

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Contact type	A	B	C	D	E	Contact ident. no.
91 92 93 94 95 96 97 98 99	110	I II III IV V VI VII VIII IX	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)	1-55
100 101 102 103 104 105 106 107 108	120	I II III IV V VI VII VIII IX	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)	1-60
109 110 111 112 113 114 115 116 117	125	I II III IV V VI VII VIII IX	7.050 (179.07)	6.750 (171.45)	6.350 (161.29)	62	6.200 (157.48)	1-63
118 119 120 121 122 123 124 125 126	130	I II III IV V VI VII VIII IX	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)	1-65
127 128 129 130 131 132 133 134 135	134	I II III IV V VI VII VIII IX	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)	1-67

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued.

Dash no.	No. of contacts	Contact type	A	B	C	D	E	Contact ident. no.
136 137 138 139 140 141 142 143 144	140	I II III IV V VI VII VIII IX	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)	1-70
145 146 147 148 149 150 151 152 153	150	I II III IV V VI VII VIII IX	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)	1-75

TABLE II. Contact plating requirements.

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area, .170 inch minimum length, shall be gold over nickel in accordance with MIL-C-55302. Compliant pin area (contact type I, .748 inch minimum length; type II, .548 minimum length; type III, .343 inch minimum length; type IV, .265 inch minimum length; type V, .192 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302. Solder post area (contact type VI, .205 minimum length; type VII, .190 inch minimum length; type VIII, .165 inch minimum length; type IX, .135 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.
C	Overall contact finish: Shall be palladium or palladium nickel alloy (86/14 ±6 percent), 50 microinches minimum thickness, with a gold flash of 3 microinches minimum, over a nickel underplate of 50 microinches minimum in accordance with QQ-N-290,and shall meet the requirements of MIL-P-45209.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, part number MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-C-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Compliant section shall meet the requirements of MIL-STD-2166, type III. For use with .093 minimum thick printed wiring boards. Sequence of tests and number of samples shall be in accordance with MIL-STD-2166, table III.

Wrappost contacts shall be in accordance with MIL-STD-1130.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .094; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part number: M55302/173 (and dash number from table I and type from table II).

Example: M55302/173-01A.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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